

1 - Creating Database

-- 1. Create and select database

```
CREATE DATABASE IF NOT EXISTS school_db;  
USE school_db;
```

-- 2. Drop existing tables to ensure a clean setup

```
DROP TABLE IF EXISTS Performance_Audit;  
DROP TABLE IF EXISTS Primary_Students;  
DROP TABLE IF EXISTS Secondary_Students;
```

-- 3. Create Primary Students table

```
CREATE TABLE Primary_Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    email VARCHAR(100),  
    marks INT,  
    grade_level VARCHAR(10)  
);
```

-- 4. Create Secondary Students table (for Set Operations)

```
CREATE TABLE Secondary_Students (  
    student_id INT PRIMARY KEY,  
    name VARCHAR(50),  
    email VARCHAR(100),  
    marks INT,  
    grade_level VARCHAR(10)  
);
```

-- 5. Insert sample data featuring Anju, Shiya, and Raghav

```
INSERT INTO Primary_Students (student_id, name, email, marks, grade_level) VALUES  
(101, 'Anju Sharma', 'anju.s@school.edu', 88, 'Grade 5'),  
(102, 'Shiya Verma', NULL, 92, 'Grade 5'),  
(103, 'Raghav Kumar', 'raghav.k@school.edu', 74, 'Grade 6');
```

```
INSERT INTO Secondary_Students (student_id, name, email, marks, grade_level) VALUES  
(104, 'Anju Sharma', 'anju.s@school.edu', 88, 'Grade 5'), -- Duplicate entry for testing UNION  
(105, 'Kavya Nair', 'kavya.n@school.edu', 95, 'Grade 7'),  
(106, 'Rohan Das', 'rohan.d@school.edu', 65, 'Grade 8');
```

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-- Query 1: Find students whose names start with 'A' or 'S'

```
SELECT student_id, name, email
FROM Primary_Students
WHERE name LIKE 'A%' OR name LIKE 'S%';
```

```
-- Query 2: Find students whose email contains 'school'
SELECT student_id, name, email
FROM Primary_Students
WHERE email LIKE '%school%';
```

Result Grid			
Filter Rows:			
Edit:			
	student_id	name	email
▶	101	Anju Sharma	anju.s@school.edu
	103	Raghav Kumar	raghav.k@school.edu
*	NULL	NULL	NULL

Primary_Students 1 Primary_Students 2 ×

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```
-- Query 1: Find students scoring between 75 and 90 marks (Inclusive)
SELECT student_id, name, marks
FROM Primary_Students
WHERE marks BETWEEN 75 AND 90;
```

```
-- Query 2: Find students whose marks fall outside the 80 to 95 range
SELECT student_id, name, marks
FROM Primary_Students
WHERE marks NOT BETWEEN 80 AND 95;
```

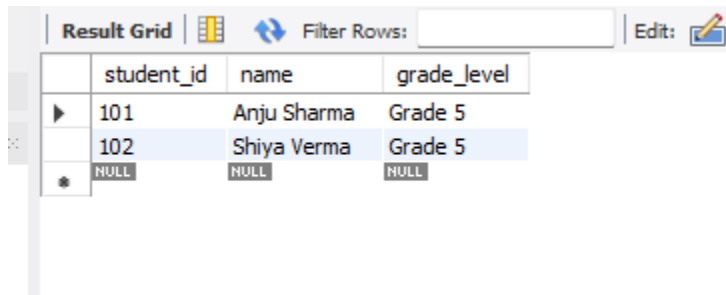
Result Grid			
Filter Rows:			
Edit:			
	student_id	name	marks
▶	103	Raghav Kumar	74
*	NULL	NULL	NULL

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```
-- Query 1: Filter students by an explicit list of target names
SELECT student_id, name, grade_level
```

```
FROM Primary_Students
WHERE name IN ('Anju Sharma', 'Raghav Kumar');
```

```
-- Query 2: Exclude specific grade levels using NOT IN
SELECT student_id, name, grade_level
FROM Primary_Students
WHERE grade_level NOT IN ('Grade 6', 'Grade 8');
```

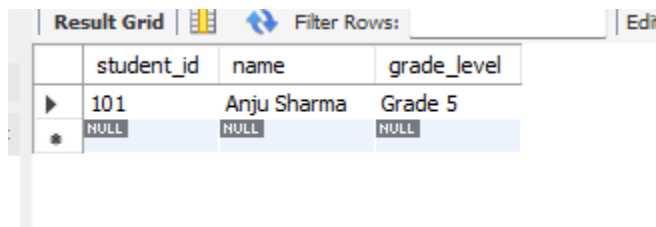


The screenshot shows a 'Result Grid' with columns: student_id, name, and grade_level. The first row contains '101', 'Anju Sharma', and 'Grade 5'. The second row contains '102', 'Shiya Verma', and 'Grade 5'. Below these, there is a row with 'NULL' values for all three columns, preceded by a star icon.

student_id	name	grade_level
101	Anju Sharma	Grade 5
102	Shiya Verma	Grade 5
NULL	NULL	NULL

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```
-- Find students in Primary_Students who also exist in Secondary_Students (by matching email)
SELECT p.student_id, p.name, p.grade_level
FROM Primary_Students p
WHERE EXISTS (
    SELECT 1
    FROM Secondary_Students s
    WHERE s.email = p.email
);
```



The screenshot shows a 'Result Grid' with columns: student_id, name, and grade_level. The first row contains '101', 'Anju Sharma', and 'Grade 5'. Below this, there is a row with 'NULL' values for all three columns, preceded by a star icon.

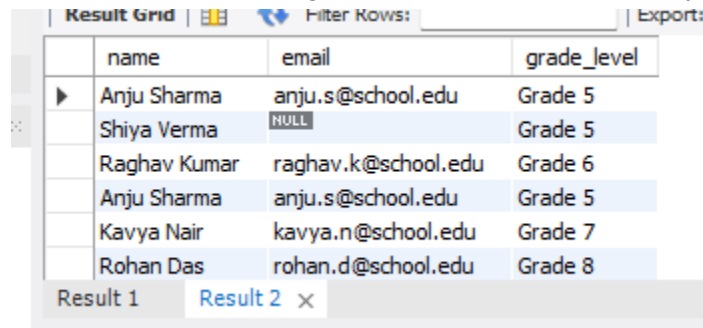
student_id	name	grade_level
101	Anju Sharma	Grade 5
NULL	NULL	NULL

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```
-- Query 1: UNION (Combines both tables and REMOVES duplicate rows)
-- Note: 'Anju Sharma' appears in both tables, but will appear only ONCE in output.
SELECT name, email, grade_level FROM Primary_Students
UNION
SELECT name, email, grade_level FROM Secondary_Students;
```

```
-- Query 2: UNION ALL (Combines both tables and RETAINS all duplicates)
-- Note: 'Anju Sharma' will appear TWICE in output.
```

```
SELECT name, email, grade_level FROM Primary_Students
UNION ALL
SELECT name, email, grade_level FROM Secondary_Students;
```



Result Grid | Filter Rows: | Export:

	name	email	grade_level
▶	Anju Sharma	anju.s@school.edu	Grade 5
	Shiya Verma	NULL	Grade 5
	Raghav Kumar	raghav.k@school.edu	Grade 6
	Anju Sharma	anju.s@school.edu	Grade 5
	Kavya Nair	kavya.n@school.edu	Grade 7
	Rohan Das	rohan.d@school.edu	Grade 8

Result 1 | Result 2 x